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Mahidol University

## INTERNSHIP Project Proposal

# Dynamic follow-up questions for attributing failures of RAG chatbot


Advisors
Asst. Prof. Dr. Thanapon Noraset
Assoc. Prof. Dr. Natthawut Kertkeidkachorn (JAIST)

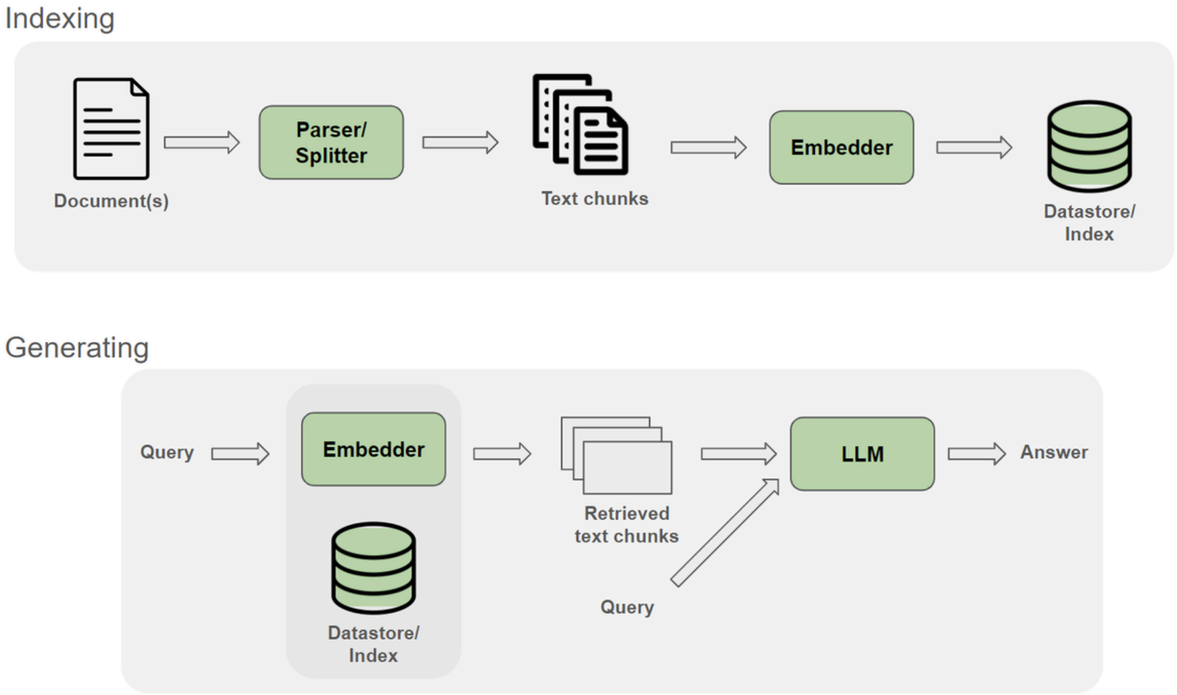
Presented By
Tanyarat Suksuwan
Kanyanat  Wasuthawat
Sippakorn Kanin

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# Introduction

# What is RAG?




Retrieval augmented generation, or RAG, is an architecture for optimizing the performance of an artificial intelligence (AI) model by connecting it with external knowledge bases. RAG helps large language models (LLMs) deliver more relevant responses at a higher quality.

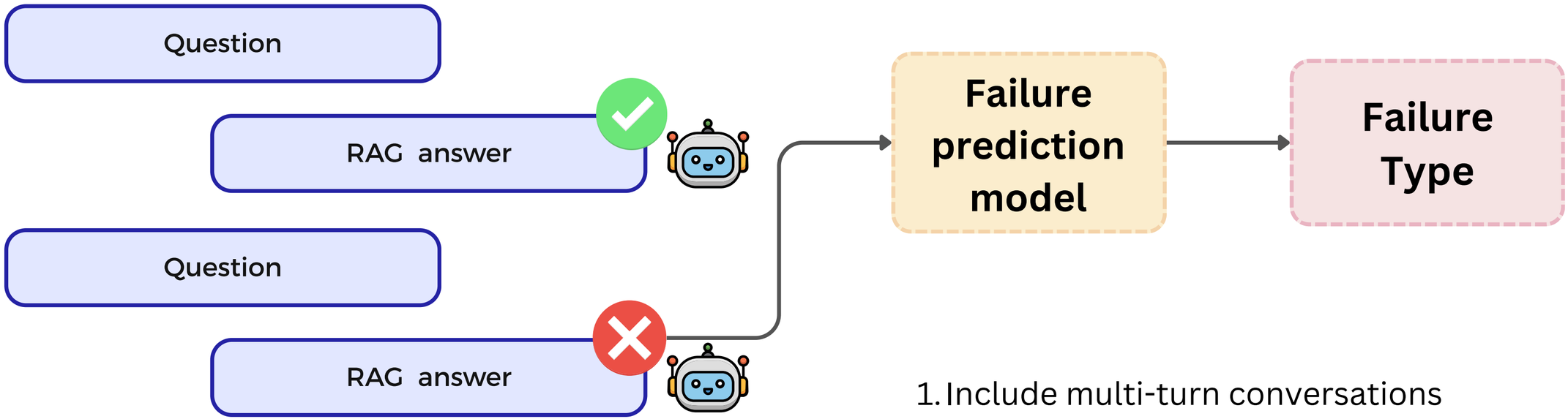
# The Problem of RAG Evaluation

RAG systems consist of multiple stages, and errors in the final answer can originate from different components.

- Current RAG evaluation can often tell that the answer is wrong. But it does not clearly tell what kind of failure caused the wrong answer.
- For example:
  - Did the corpus not contain the answer?
  - Was the evidence badly chunked?
  - Did retrieval miss the right chunk?
  - Was the right chunk retrieved but not selected?
  - Did the model answer only part of the question?

# Scope of the Project

To find the more method that can attribute failure in RAG Conversation

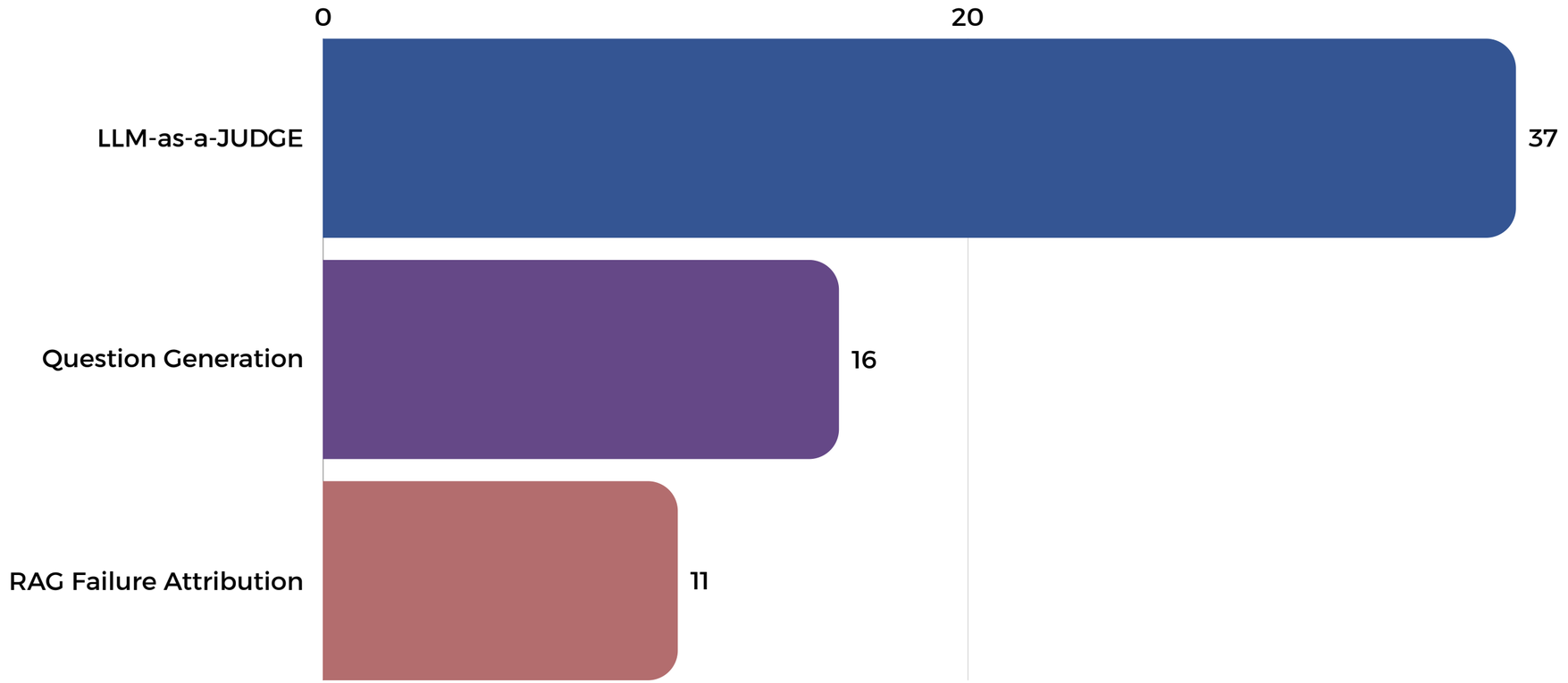



1. Include multi-turn conversations
2. Generalize to many datasets
3. Focus on the English language

# Literature Review

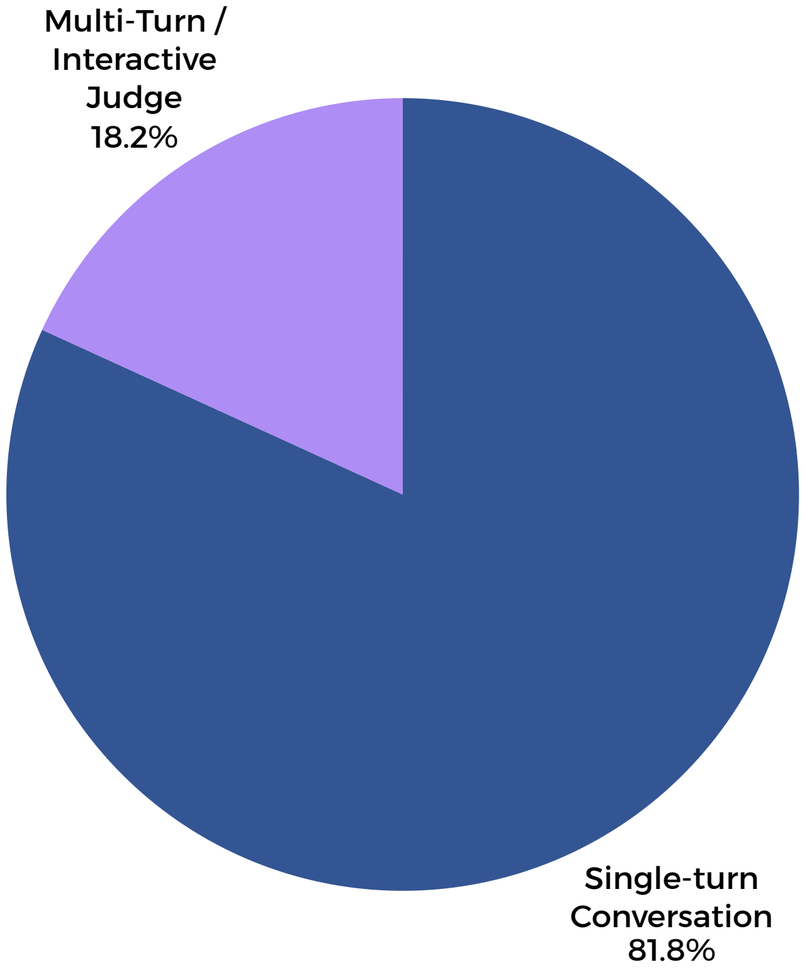
# Literature Review

We reviewed 63 papers for chatbot evaluation and found 3 main groups of related papers:




LLM-as-a-judge

# Literature Review: LLM-as-a-Judge
# Frequency Multi-turn vs Single-turn




## There are just 18.2% of multi-turn from reviewing research paper

**Multi-turn:** Evaluates the RAG system across continuous, back-and-forth dialogues, accounting for context shifts, conversational dependencies, and cumulative error propagation over time.

List of multi-turn/interactive judge papers:

- SLEF-RAG
- MT-Bench
- RAD-Bench: Evaluating LLM Capabilities in Retrieval Augmented Dialogues
- Let the LLMs Talk: Simulating Human-to-Human Conversational QA
- The 2nd FutureDial Challenge: Dialog Systems with RAG
- Meta-Rewarding Language Models: Self-Improving Alignment with LLM-as-a-Meta-Judge

# MT-Bench-101: A Fine-Grained Benchmark for Evaluating Large Language Models in Multi-Turn Dialogues

The AI assistant's general reasoning capabilities are crucial for accurately addressing and explaining a wide range of problems posed by 'Human'. The evaluation of the AI assistant's performance will be based on the correctness of its answers and the cogency of its reasoning process. The evaluation criteria are as follows:

1. Verify the accuracy of the AI assistant's answer against the provided reference solution in the format '### reference solution ###' for the specific problem.
2. Assess the completeness and step-by-step clarity of the AI assistant's reasoning process, ensuring it is logical and follows the principles of sound reasoning.
3. Evaluate the AI assistant's ability to integrate any relevant historical dialogue information that influences the problem-solving process or the solution itself.
4. Appraise the AI assistant's communication of the solution in a manner that is understandable and instructive to 'Human', potentially aiding their learning or comprehension.

Scoring Guidelines:

1-3 points: The AI assistant provides incorrect answers and/or fails to offer a clear and logical reasoning process, missing key steps or providing explanations that do not adhere to standards of sound reasoning.
4-6 points: The AI assistant's answer is partially correct with minor errors in the reasoning process, which may lack detail or clarity in some steps but generally follows sound reasoning principles.
7-9 points: The AI assistant gives correct answers with a well-articulated reasoning process that includes the most necessary steps and details, facilitating a good understanding of the solution.
10 points: The AI assistant provides a completely correct answer accompanied by a detailed and meticulously clear step-by-step reasoning process that is fully aligned with sound reasoning principles and enhances 'Human's understanding.
When scoring, focus on the precision of the AI assistant's answer and the extent to which the reasoning process is elaborated. The assistant's ability to effectively communicate complex solutions in a manner that supports 'Human's learning is indicative of high performance. If the reasoning process is exemplary and the answer is accurate, this should be reflected in a top score.

Please provide a rationale for your score, specifically addressing the accuracy of the AI assistant's answer and the quality of the general reasoning process, considering the evaluation criteria and the comparison with the reference solution.

Figure 32: The evaluation prompt for general reasoning task.




- Uses an LLM (e.g., GPT-4) as the judge with a rubric and scoring prompt
- Evaluates multi-turn conversations
  - The judge receives the conversation context, including the previous turns, and then evaluates the model's responses.
- Compares LLM judges with human preferences
  - Instruction following
  - Correctness
  - Relevance
  - Helpfulness
  - Coherence

# Literature Review: LLM-as-a-Judge
# Main Judge Criteria (1/2)

| Criteria | Measures |
|---|---|
| Helpfulness / relevance: | does the answer solve the user's question or task? |
| Factuality / accuracy / faithfulness: | is it correct and grounded in the given context? |
| Citation / evidence support | does it use and cite the provided evidence correctly? |
| Reasoning / logic | can it explain steps, handle multi-hop logic, and avoid weak reasoning? |
| Completeness / coverage | does it include all important information or atomic facts? |

# Literature Review: LLM-as-a-Judge
# Main Judge Criteria (2/2)

| Criteria | Measures |
|---|---|
| Consistency / coherence / conflict | is the answer stable, clear, and not contradictory? |
| Safety / harmlessness | does it avoid harmful, unsafe, or policy-violating content? |
| Abstain / negative rejection | can it say “not enough information” when needed? |
| Robustness / bias sensitivity | is the judge resistant to position bias, verbosity, fake citations, and style artifacts? |
| Efficiency / interpretability | is the judgment transparent and not overthinking simple tasks? |

# Literature Review: LLM-as-a-Judge
# Takeaway Messages

- **Multi-Turn / Interactive Judge:**
  - Most papers study single-turn evaluation
  - Most RAG is deployed in a multi-turn environment

- **Judge criteria:**
  - Measuring the correctness and alignment of the answer with human expectations
  - Not attributing the root cause of RAG for improvement.

# Literature Review: Question Generation
# Types of Generated Questions (1/3)

| Type | Description |
| --- | --- |
| Fact Question (Single-Hop) | The answer is found in exactly one sentence or document. |
| Multi-Hop / Cross-Document | The model must connect clues scattered across multiple documents to find the answer. |
| Information Integration | Merging separate facts from multiple sources into a single, cohesive answer. |
| Negative Rejection / Appropriate Abstain | Reject when the retrieved text doesn't contain the answer, instead of hallucinating. |

# Literature Review: Question Generation
# Types of Generated Questions (2/3)

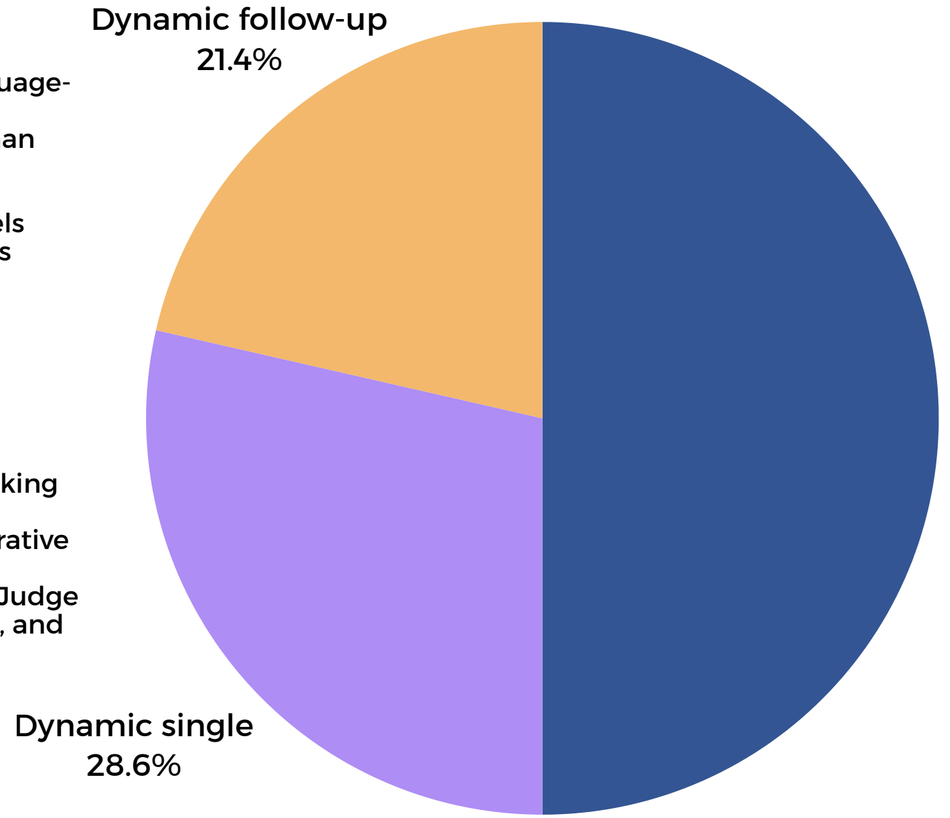
| Type | Description |
| --- | --- |
| Noise Robustness | Testing how well the model filters out irrelevant filler text to find the true answer. |
| Counterfactual / Conflict Robustness | Checks if it trusts the context. (error detect) |
| Open-End | Broad questions requiring long-form, natural language generation instead of short answers. |
| Multi-Turn Dialogue | A back-and-forth chat simulation where queries change based on history. |

# Literature Review: Question Generation
# Types of Generated Questions (3/3)

| Type | Description |
| --- | --- |
| Fine-Grained Citation | Forcing the system to tag exactly which line or document it pulled its answer from. |
| Domain-Specific (Medical / Product QA) | Questions testing specialized, highly technical industry leaflets or corporate datasheets. |
| Knowledge-Intensive Tasks (KILT / SuperGLUE) | Massive academic datasets used to test deep open-domain knowledge. |
| General Instruction Following | Platforms that grade models based on how well they obey formatting and style constraints. |

# Literature Review: Question Generation Types of Methods to generate Questions

- Benchmarking Foundation Models with Language-Model-as-an-Examiner
- Let the LLMs Talk: Simulating Human-to-Human Conversational QA via Zero-Shot LLM-to-LLM Interactions
- RAD-Bench: Evaluating Large Language Models Capabilities in Retrieval Augmented Dialogues

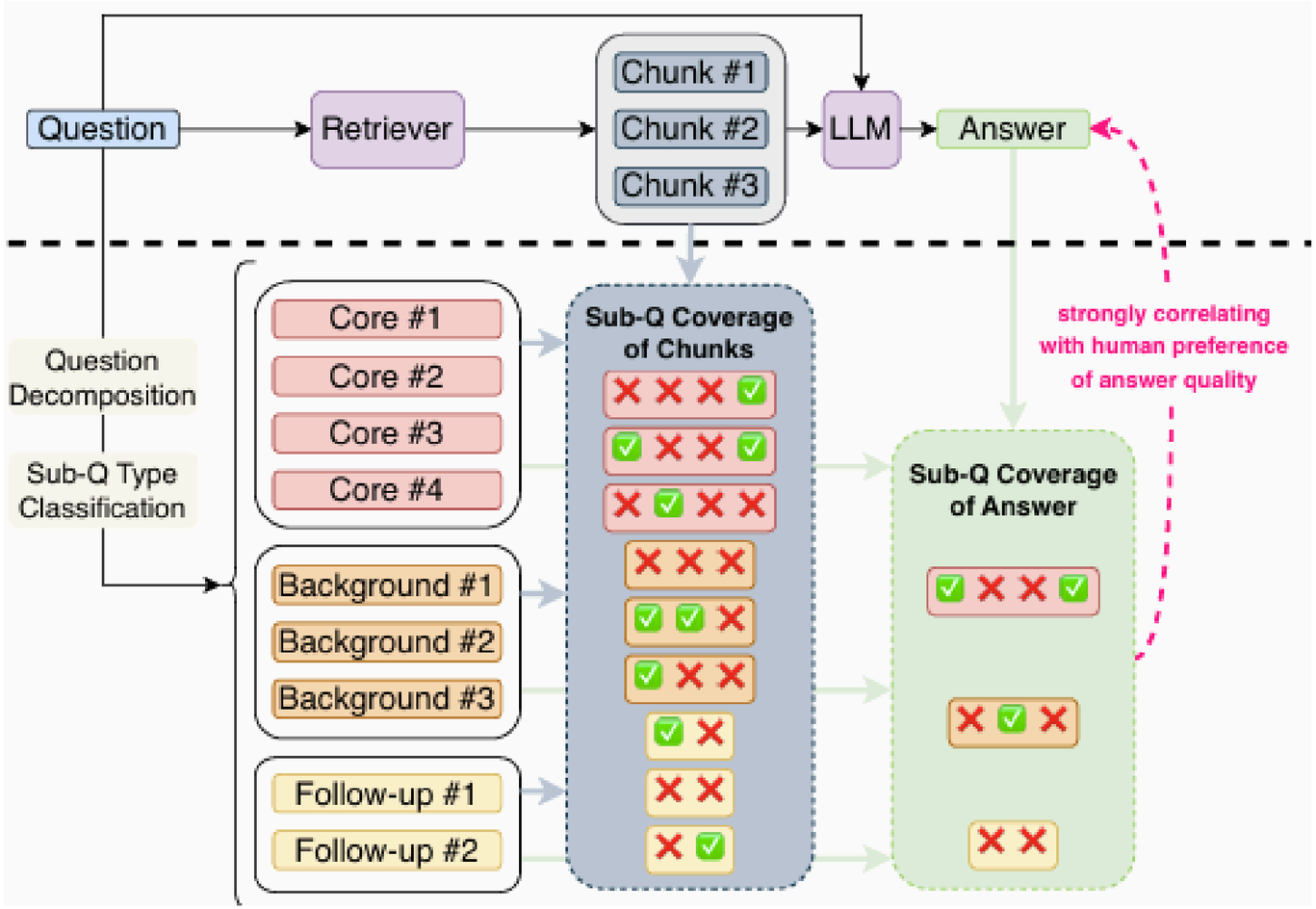



- Can LLMs Improve Multimodal Fact-Checking by Asking Relevant Questions?
- Retrieving Supporting Evidence for Generative Question Answering
- JudgeLRM: Large Reasoning Models as a Judge
- SELF-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection

- HopWeaver: Cross-Document Synthesis of High-Quality and Authentic Multi-Hop Questions
- Automated Question-Answer Generation for Evaluating RAG-based Chatbots
- Ragas: Automated Evaluation of Retrieval Augmented Generation
- ARES: An Automated Evaluation Framework for Retrieval-Augmented Generation Systems
- The Great Nugget Recall: Automating Fact Extraction and RAG Evaluation
- Benchmarking Large Language Models in Retrieval-Augmented Generation (RGB)
- Test of Time: A Benchmark for Evaluating LLMs on Temporal Reasoning

# Do RAG Systems Cover What Matters? Evaluating and Optimizing Responses with Sub-Question Coverage

**(Xie et al., 2025)**



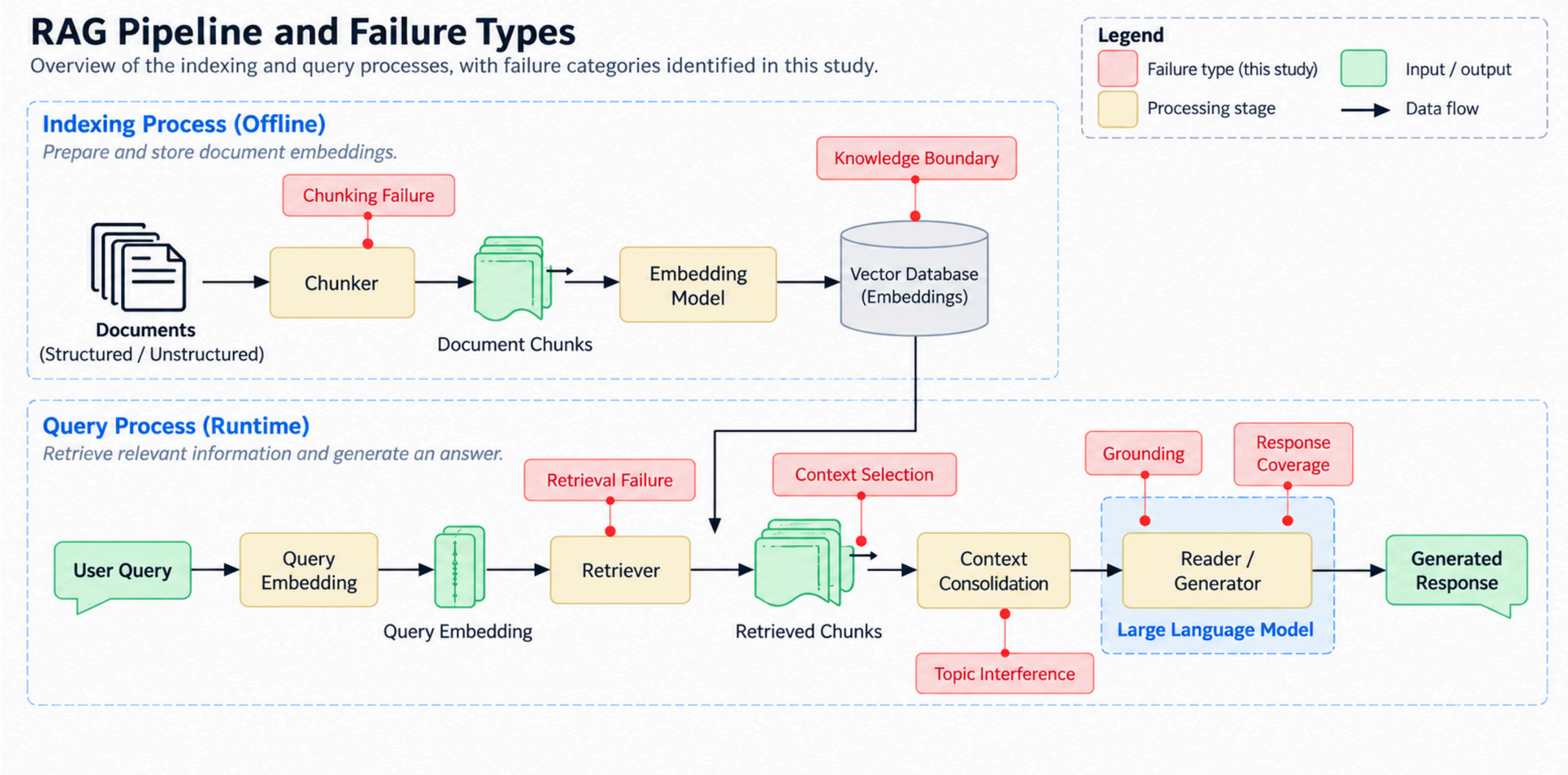

**Summary:**

- Generate sub-questions to **gain insight into RAG behavior**
- **static single turn**
- The method is helpful in improving LLM-as-a-Judge evaluation
- Not directly attributing failure types

# Literature Review: Question Generation Takeaway Messages

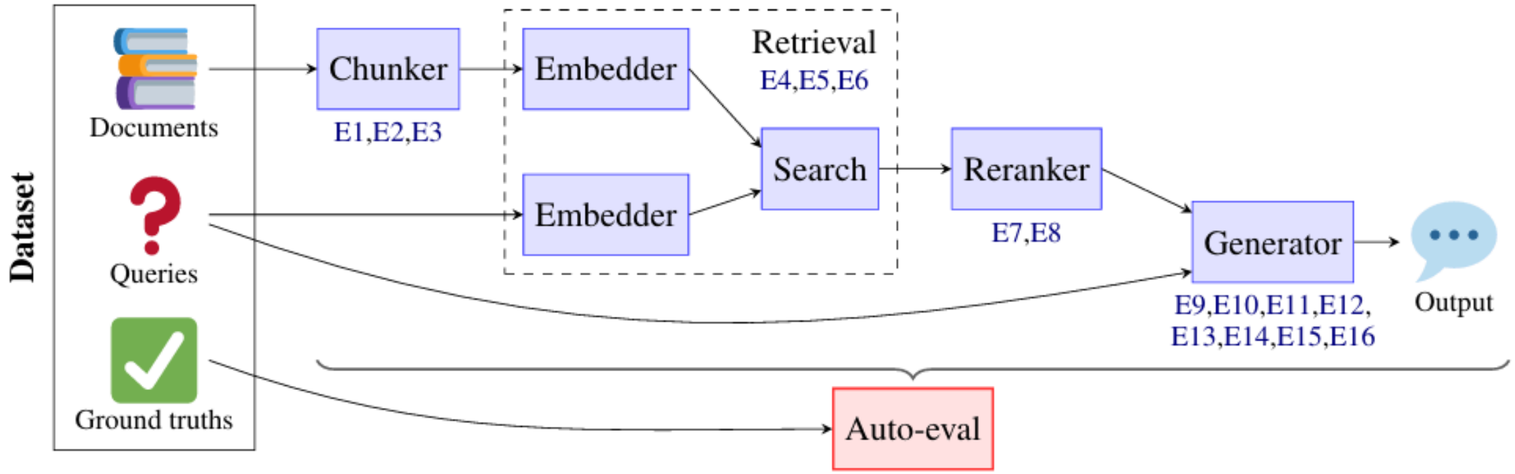
- **Many types of questions:**
  - Many types of questions aim to evaluate RAG, **not probe** RAG failures
  - Many types can be adopted, e.g.,
    - Fact Question
    - Clarification
    - Multi-hop
    - Information Integration
    - Counterfactual
- **Potential of Dynamic Follow-up:**
  - Probing questions should depend on the conversation (dynamic, not static)
  - Asking additional questions might reveal more information (follow-up, not single)

# Literature Review: RAG Failure Types
# Types of Failures



# Classifying and Addressing the Diversity of Errors in Retrieval-Augmented Generation Systems (DragonBall)

**(Leung et al., 2026)**




**Summary:**

- Analyze each category of failure for each RAG stage
- Human annotation of RAG failures
- Single-turn conversation
- Only use conversation trace, no additional probing questions

```
"""
You are an expert evaluator for
    Retrieval Augmented Generation (RAG)
     systems. You will help identify
    chunking-related errors in RAG
    systems by analyzing the
    relationship between queries, ground
     truth answers, and document chunks.
     Your task is to identify which
    chunks from the ground truth
    document are relevant to answering
    the query.

## Instructions
1. Review ALL available chunks from the
    ground truth document
2. Identify ALL chunks containing
    information relevant to answering
    the query
3. Consider both direct and indirect
    relevance to the query
4. Select chunks that together provide
    sufficient information to answer the
     query

## Context
**Query**: {query}
**Ground Truth Answer**: {ground_truth}

**Available Chunks from Ground Truth
    Document**:
{chunks}

## Output Format
Provide your answer in the following
    format:
Relevant Chunks: [45_1, 45_4, 45_10]

"""
```

# Literature Review: RAG Failure Types
# Types of Failures

We found that Leung et al., 2025 failures can be applied to mult-turn conversation as well.

| Failure Types | Description | Single-Turn | Multi-Turn |
|---|---|---|---|
| Knowledge Boundary | The system fails to recognize that the answer is outside the available evidence scope. | The corpus does not contain enough evidence, but the system answers anyway. | The system cannot tell whether the answer should come from retrieved evidence, prior conversation, or is unsupported by both. |
| Chunking | Relevant information exists in the source collection, but the chunk structure makes it hard to retrieve or use. | Needed evidence is split, incomplete, too broad, too narrow, or lacks surrounding context. | The evidence needed may span what the user has built up over several turns; chunking fails when no chunk holds a self-contained answer to the (often under-specified or follow-up) question, or when sub-answers needed across turns are scattered. |
| Retrieval | The system fails to retrieve relevant evidence that exists in the indexed corpus. | A clear standalone query has relevant chunks available, but they are not retrieved. | The follow-up query is underspecified or context-dependent, causing retrieval for the wrong entity, topic, or aspect. |
| Context Selection | Relevant evidence is available after retrieval but is not selected for generation. | Relevant chunks are retrieved but dropped, ranked too low, or replaced by distractors. | The system fails to select the right evidence across turns, especially when the answer needs both current-turn evidence and earlier conversational context. |
| Grounding | The answer is inconsistent with, unsupported by, or overextends beyond the selected evidence. | The model fabricates, contradicts evidence, or makes claims not supported by the context. | The model contradicts retrieved evidence or established conversation state, mixes entities across turns, or imports unsupported prior assumptions. |
| Response Coverage | The answer is supported but incomplete relative to the user's request. | The answer omits part of a multi-part question, misses key evidence, or gives only a partial answer. | The answer addresses the latest turn but misses accumulated constraints, comparisons, requested scope, or unresolved pieces from earlier turns. |

# Research Gap

# Literature Review: Research Gaps

- Most work focuses on LLM-as-a-Judge to evaluate RAG
- There is a framework for single-turn RAG failure type classification (Leung et al., 2026)
- Researchers apply probing questions to investigate RAG failures (Xie et al., 2026)

# Research Question

## Research Question:

## Can Dynamic follow-up question attribute failures of RAG more accurately than static sub-question generation?

**Project Objectives:**

- To develop and evaluate a **Dynamic Follow-up Question** framework that uses runtime execution traces and **multi-turn** context shifts
- To attribute **RAG pipeline failures** (such as retrieval misses, context selection errors, and grounding failures) significantly more accurately than static sub-question generation methods.

# Research Methodology

# Research Methodology

**Data Collection 1: Development Set**

Multi-turn
MT-RAG
x 110 convos

Single-turn Q&A
dragon ball
x 400 convos

y - Human Annotation

**Data Collection 2: Held-out Set**

Judge

Correct

Y

N

End

**X:**

**Method B0: No Probing Questions**

Q-A log

Question

Answer

Question

Retrieve Chunks

1

2

Ground Truth Answer, Chunk

Answer1:

Answer2:

**Method B0 + B1: Xie's sub questions**

Xie's Judgement 1

Xie's Judgement 2

**Method B0 + B2: Dynamic Follow-up Questions**

Q-A log + Chunks

Q-type

Q-quality

Failure Predicted Model

ŷ - Failure type

y - Human Annotation

Evaluation
(Accuracy, F1)

Dev Set

Test Set

**POC: Dynamic Follow-up**

Evaluate Question Quality

# Methodology

## Data Collection

### Data Collection 1: Development Set



**Single-turn Q&A**
**dragonball**

- Human-annotated failure types
- 400 conversations with standard RAG (LLaMA-3 8B and GTE Large English)
- We reproduced the retrieve chunk.



**Multi-turn**
**MT-RAG**

- No existing human-annotated failure types
- 200 human-reviewed multi-turn conversations
- 4 sub-datasets → CLAPnq, FiQA, Govt, Cloud

- We run 4 RAG systems: RAPTOR, Self-RAG, HippoRAG, and standard RAG to collect a RAG conversation dataset
- We annotate failure type for these conversations.

### Data Collection 2: Held-out Set

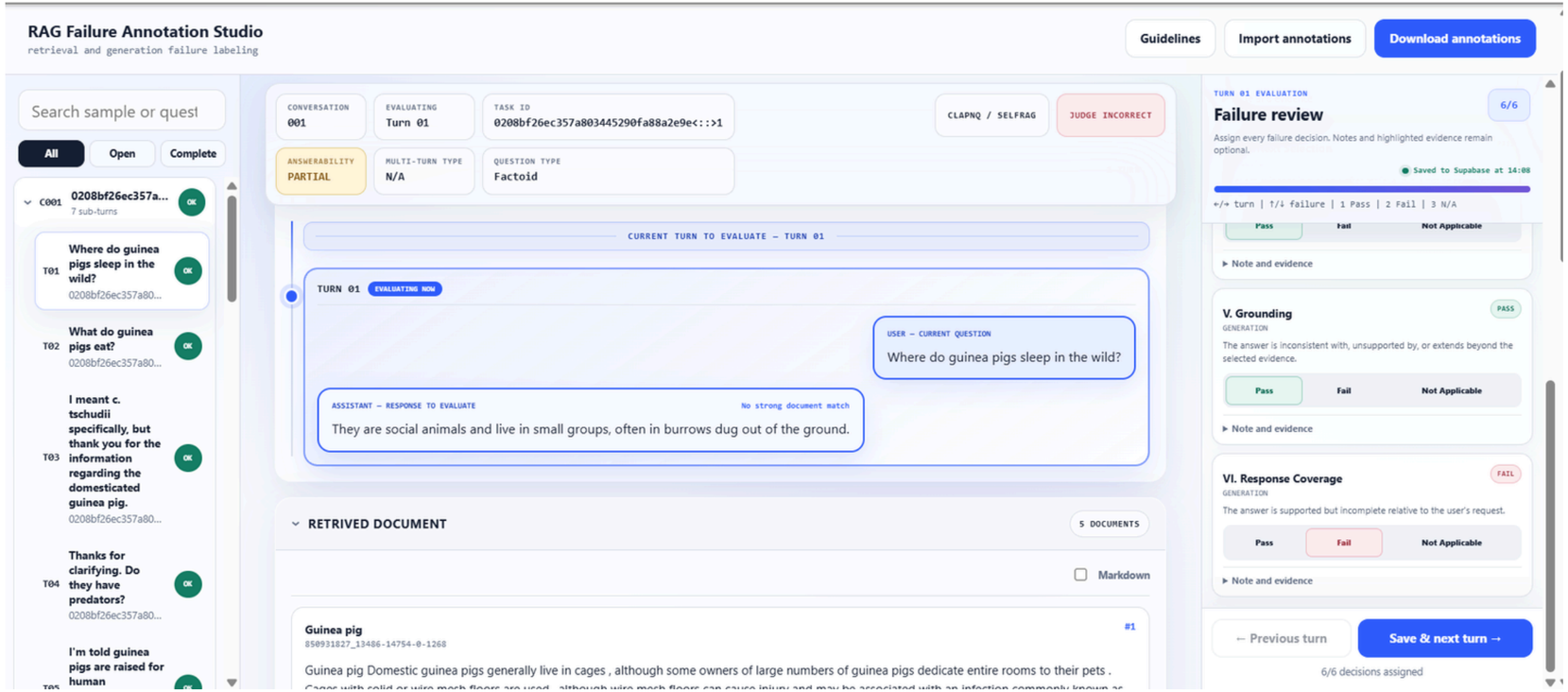
- 4-5 Different Datasets
- We plan to sample a few conversations from many existing datasets to test the generalization across diverse domains.

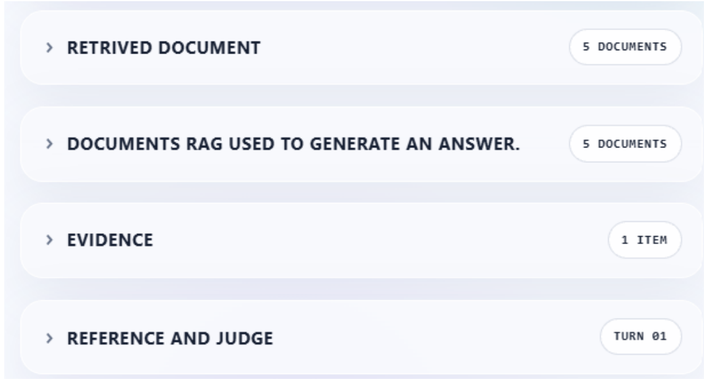


# Methodology

## Data Collection: Human Annotation (y) Method

Create Human Annotaion guideline and annotate by 3 annotators




**Procedure**

- Separate annotate and compare the each to find cohesive
- Annotate each failure turn

# Methodology

## Three question Generation Methods

| Method | Input | Generation Strategy | output to annotate (X) |
|---|---|---|---|
| **Method B0:** No Additional | • Failed Q-A log<br>• Retrieved chunks<br>• Ground truth answer & chunk | None (0 additional questions generated) | Q-A log + Chunks + Ground truth |
| **Method B0+B1:** Xie's Sub-Question | • Failed Q-A log<br>• Retrieved chunks<br>• Ground truth answer & chunk | Breaks the original query into sub-questions(core, background, follow-up) and evaluates.<br>• Checks retrieve document: Does the retrieved text contain the facts needed to answer?<br>• Checks Answer: Does the final RAG response actually include those facts? | Q-A log + Chunks + Ground truth + Coverage chunk + Coverage answer |
| **Method B0+B2:** Dynamic Follow-up | • Failed Q-A log<br>• Retrieved chunks<br>• Gold answer<br>• Ground truth answer & chunk | Generates dynamic follow-ups conditioned on how RAG answered | Q-A log + Chunks + Ground truth + Dynamic Q-A conversation logs |

# Approach to generate dynamic follow-up questions

## Inspire by paper LLM-AS-AN-INTERVIEWER: Beyond Static Testing Through Dynamic LLM Evaluation (Kim et al., 2026)

| Follow-Up Question | TYPE and Example |
|---|---|
| Knowledge/Factuality | ADDITIONAL FACTS **Can you provide an example** of determining the 6th roots of unity and specifying their arguments in radians? |
| Reasoning | CLARIFICATION **What does it mean** for $a$ and $b$ to be consecutive integers? |
| | RATIONALE **How did you determine** Jasmine's rate of water consumption per mile? |
| Instruction Following | MODIFICATION OF CONDITIONS **If we change the previous question** and assume that each sister of David has two brothers, how many brothers would David have? |
| | ADDITIONAL EXPLANATION **Can you provide a specific example of how you would address** a potential objection your friend might have about public speaking in your persuasive email? |
| | CORRECTION **Can you now reformulate** your earlier reply, outputting it in JSON format and only including books published after 1980? |

**D.2.2 Follow-Up Question Generation**

**MATH Fail:**
If the seed question is answered incorrectly, generate a clarification question based on the model's error type. The question should focus on the concept or interpretation of the seed question to help identify the source of the misunderstanding.
**MATH Success:**
If the seed question is answered correctly, generate a follow-up rationale question to assess the model's understanding of the reasoning behind its solution.
**DepthQA:** Regardless of whether the seed question is answered correctly or not, generate a follow-up question based on the reference material. The question should inquire about additional facts or details that the model did not address in its response.

MATH Fail

**Task:**
You are a question generator for assessing the model. Below are the list of the answer history of an Evaluatee model that keeps getting the question wrong, its error type, the corresponding problem, and the solution. Your role is to generate a proper follow-up question based on the model's error type.
**Instructions:** 1. Model's Error Type is `{error_type}`. Do not disclose the error type to the model.
2. Based on the error type, generate appropriate feedback as described below.
**Possible Error Types:**
1. **Concept:** This error type indicates the model lacks the concept used to solve the problem. The feedback should contain a question to assess the understanding of the concept. (e.g., What is the Pythagorean theorem?)
2. **Misinterpret:** The model misunderstood or misinterpreted the question. The feedback should include a follow-up question or clarification to help the model reassess and correctly interpret the original question. This ensures the model understands the context and requirements before proceeding.
3. **Calculation:** The model made a mistake in calculation. Feedback should focus on identifying the part that requires recalculation.
4. **N/A:** None of the above. Provide a custom description.
**Response Format (JSON):** {Format Example}
**Input Data:**
Question: {question}
Correct Answer (DO NOT disclose): {answer}
Reference Solution (DO NOT disclose): {solution}
Previous Feedback: {Dialogue_History}
Model's Error Type: {error_type}

**Output:**

Probe how well the model understands/defends its own answer, and reduce judge bias/contamination and already assigns error_type

**Adapt**

- **Build LLM based signal from retrieved chunks, gold chunk, gold answer instead asssign the error_type**
- **Using MATH Fail prompt and swapping their 4 math error types for our 6 RAG failure type**

# Methodology

## Judge: Verification Prompt Logic

### Single-Turn

A prompt with Q-A log judges

**If Correct:** pipeline ends immediately. No failure attribution required.

**If Incorrect:** The Q-A log is forwarded to the next step (Question Generation & Failure Prediction).

### Multi-Turn

Turn-by-turn prompt judging across multi-turn conversational

**If ALL Turns are Correct:** The conversation is completely successful and the pipeline ends

**If ANY Turn is Incorrect:** The entire multi-turn Q-A log is flagged and sent toward the next step.

This means one multi-turn conversation can have multiple failure annotations

# Methodology

## Failure Prediction

- **Model Input (Output X):**
  - Ingests the base failed Q-A log, retrieved chunks, ground truth, and question generation output (B0, B1, or B2).
- **Zero-shot Prediction using LLMs** (following **DragonBall):**
  - Uses DragonBall failure prompts to develop our prompt for failure prediction.
- **Model Output (ŷ):**
  - Classifies failure into 1 of the 6 failure types: Knowledge boundary, Chunking, Retrieval, Context Selection, Grounding, or Response Coverage..

```
"""
You are an expert evaluator for
    Retrieval Augmented Generation (RAG)
     systems. You will help identify
    chunking-related errors in RAG
    systems by analyzing the
    relationship between queries, ground
     truth answers, and document chunks.
     Your task is to identify which
    chunks from the ground truth
    document are relevant to answering
    the query.

## Instructions
1. Review ALL available chunks from the
    ground truth document
2. Identify ALL chunks containing
    information relevant to answering
    the query
3. Consider both direct and indirect
    relevance to the query
4. Select chunks that together provide
    sufficient information to answer the
     query

## Context
**Query**: {query}
**Ground Truth Answer**: {ground_truth}

**Available Chunks from Ground Truth
    Document**:
{chunks}

## Output Format
Provide your answer in the following
    format:
Relevant Chunks: [45_1, 45_4, 45_10]

"""
```

# Methodology

## Evaluation Metrics

$\hat{y}$ - Failure type by Classification Model

y - Failure type by Human

**Accuracy and F1 scores**

**Then Compare all methods (B0 vs B1 vs B2)**

# Preliminary Results

# Preliminary Results

## DragonBall Reproducing:

| Metric | Count | Percentage / basis |
|---|---|---|
| Total cases | 377 | 100.0% of all cases |
| Judged correct | 62 | 16.4% of all cases |
| Possibly correct | 15 | 4.0% of all cases |
| Judged wrong | 300 | 79.6% of all cases |
| Failure type classification accuracy | 117 / 300 | 39.0% of judged wrong |
| Failure stage classification accuracy | 173 / 300 | 57.7% of judged wrong |

For *error type classification*, we query the LLM $K = 10$ times for each datapoint and extract the most and second-most frequently predicted error types. Compared to human-annotated error types, RAGEC achieves an accuracy of 40.3%. Figure 2 shows the distribution of the modal error type for DragonBall-EN and CLAPnq. Table 2 shows more details for the DragonBall-EN dataset (CLAPnq is shown in Table 6 of Section C.1). These distributions differ, showing that the nature of the dataset

For *stage classification*, RAGEC achieves a 57.8% agreement rate with human annotations. Details of the results are shown in Table 1. We find that human annotations more often identify the retrieval stage as causing errors, whereas RAGEC favors blaming the generator. While RAGEC's agreement rate is not close to perfect, we tested several other methods to engineer the context for LLM-based auto-evaluation that all performed worse (described in Section B.4). Hence, RAG error classifi-

# Preliminary Results

## MT-RAG Turn that marked as incorrect

| dataset | rag | total | failures | fail rate |
|---|---|---|---|---|
| clapnq | selfrag | 224 | 172 | 76.80% |
| clapnq | vector | 224 | 79 | 35.30% |
| cloud | hipporag | 205 | 120 | 58.50% |
| cloud | selfrag | 205 | 189 | 92.20% |
| cloud | vector | 205 | 119 | 58.00% |
| fiqa | hipporag | 199 | 132 | 66.30% |
| fiqa | raptor | 199 | 105 | 52.80% |
| fiqa | selfrag | 199 | 159 | 79.90% |
| fiqa | vector | 199 | 101 | 50.80% |
| govt | hipporag | 214 | 117 | 54.70% |
| govt | selfrag | 214 | 182 | 85.00% |
| govt | vector | 214 | 112 | 52.30% |

*llama 6B embedding Contriever-MS MARCO (dense)*

*We decide to exclude selfrag, total annotation require 885 turns.*
*We will use selfrag to develop annotation guideline.*

# Preliminary Results

## Annotation Progress:

Finding the cohesive with B0 method with RAG ClapNQ dataset.by 3 annotators. Using pair-wise conhen's kappa

**Observed agreement ($p_o$)**

$$p_o = \frac{\text{number of items where both annotators gave the same label}}{\text{total number of items}}$$

**Expected agreement by chance ($p_e$):**
how often would they land on the same label just by luck

$$p_e = \sum_l \left(\frac{\text{row total for label } l}{n}\right) \times \left(\frac{\text{column total for label } l}{n}\right)$$

**Cohen's Kappa (κ)**

$$\kappa = \frac{p_o - p_e}{1 - p_e}$$

18 Failure turns from 3 conversations from selfrag clapnq dataset

| Conversation | annotator 1 vs 2 | annotator 1 vs 3 | annotator 2 vs 3 |
|---|---|---|---|
| 1 | 0.178 | 0.172 | 0.421 |
| 2 | 0.037 | 0.123 | 0.114 |
| 3 | 0.023 | 0.064 | 0.201 |

# Preliminary Results

## xie sub question example from multi-turn conversation:

**Question: who takes photos of planes in the air**
**RAG answer:** Aviation photographers
**Number of sub-questions:** 20
**Core sub-questions:** 1
**Follow-up sub-questions:** 10
**Background sub-questions:** 9
**Question in Core sub-questions: How do military photographers capture images of military aircraft in flight?**

**Example of sub-queston:**
True: What are the different types of aviation photography?
False: Who are professional aviation photographers?
True: What equipment is used by photographers who take photos of planes in the air?
True: What techniques do photographers use to capture images of planes in flight?
False: Are there specific companies that specialize in aerial photography of aircraft?
False: Do airlines employ photographers to take pictures of their aircraft?
True: How do military photographers capture images of military aircraft in flight?
False: What role do hobbyists play in aviation photography?
False: What are the legal considerations for taking photographs of planes in the air?
False: How do photographers gain access to airfields or secure areas for shooting?
True: What events or airshows are popular for photographing planes in the air?
False: How do photographers ensure safety while taking photos of planes in the air?
True: What are the challenges faced by photographers when shooting planes in the air?
False: How has drone technology impacted the photography of planes in the air?
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# PLAN

# Plan

1. **[Sep 2026] Develop Dynamic follow-up Question Generation**

2. **[Sep-Oct 2026] Collecting Human Annotation**
   - Finalize the annotation guideline
   - Annotate 885 Failure turns from MTRAG

3. **[Oct-Nov 2026] Collecting LLM Failure prediction for B0, B1, B2**
   - [Oct 2026] Single-Turn: Dragonball(Leung et al., 2026)
   - [Nov 2026] Multi-Turn: MTRAG + Our annotated dataset

4. **[Nov-Dec 2026] Improve dynamic follow up question generation**

5. **[Dec 2026] Comparing and Analyzing Results**

# Thank You

